

Rational expressions



1

a $\frac{7v}{5}$

b $\frac{3}{4}$

c $\frac{7v}{71}$

d $\frac{7x}{3}$

e $2p^4$

f $\frac{10n}{3m}$

g $\frac{-2k}{3}$

h $-\frac{y}{4x}$

2

a i $\frac{4a}{4a(b-5c)}$

ii $= 4a$

iii $\frac{1}{b-5c}$

b i $\frac{2f(g-3h)}{2f}$

ii $2f$

iii $g-3h$

c i $\frac{3r(2+st)}{3r(s-4t)}$

ii $3r$

iii $\frac{2+st}{s-4t}$

d i $\frac{4xy(1+2y)}{12x^3y^3}$

ii $4xy$

iii $\frac{1+2y}{3x^2y^2}$

3

a $\frac{3x-8}{6}$

b -1

c -3

d $\frac{x-4}{3x+5}$

e $v-4w$

f $\frac{1}{v-5w}$

g 2

h 3

4

a $-(a+9)$

b $\frac{1}{x+5}$

c $\frac{1}{x+6}$

d $m-1$

e $m+10$

f $\frac{5m+7n}{8m}$

g $\frac{1}{x-1}$

h $\frac{1}{x+6}$

i $\frac{6}{5(x+12)}$

j $x+5$

k $\frac{x+2}{10}$

l $\frac{2(x+11)}{5}$

5

a $\frac{2}{r+4}$

b $\frac{2}{x}$

c $-\frac{5}{2x}$

d $\frac{5}{x^2}$

e $\frac{4}{r-5}$

f $\frac{p-q}{p+q}$

g $-\frac{3}{x}$

h $\frac{1}{3}$

6

a $\frac{x+4}{x-5}$

e $\frac{x-7}{x+7}$

i $\frac{2x+3}{3-2x}$

b $\frac{x-7}{x-8}$

f $\frac{x+4}{x}$

j $-\frac{4x+5}{2+x}$



c $\frac{2x+3}{4x-5}$

g $\frac{a+6}{a-4}$

k $-\frac{x+4}{2(2+x)}$

d $\frac{3x+2}{6x-5}$

h $\frac{p+20}{p-11}$

l $\frac{x+2}{x(x+4)}$

7

a $x-5$

e $\frac{4(x+4)}{x+5}$

i $\frac{2(4x-5)}{4x+5}$

b $\frac{9x}{10(x+1)}$

f $\frac{x(x-5)}{x+3}$

j $\frac{2x-1}{4x-5}$

c $\frac{x+7}{x-3}$

g $-\frac{5+4x}{x+3}$

k $\frac{x-5}{x+1}$

d $\frac{x-5}{x+10}$

h $\frac{x-3}{5x-2}$

l $\frac{4x-1}{3-x}$

8 Luciana made an error in Step 2. She cannot divide out $2x$ because it is not a factor of the numerator. In fact, the result of Step 1 is already fully simplified.

9 Qin made an error in Step 3. The additive property of fractions is $\frac{A+B}{C} = \frac{A}{C} + \frac{B}{C}$, but instead Qin tried to use $\frac{A}{B+C} = \frac{A}{B} + \frac{A}{C}$ which is not true. In fact, the result of Step 2 is already fully simplified.

10

a k can be any multiple of 4.c $k = 0$ b $k = 13$ d $k = 2$ and $k = -9$